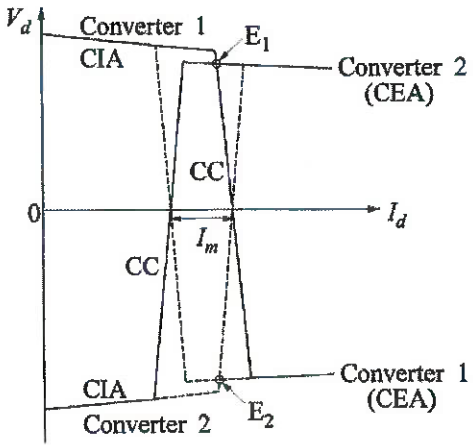
Answer ALL Questions

Question 1 and 2 carries 25 marks each, Question 3 carries 30 marks

1. a) Power reversal in an LCC HVDC link is achieved through reversal of voltage polarity. Sketch the combined rectifier-inverter characteristics of the converters at both ends and describe how the power flow through an LCC HVDC link is actually reversed in practice.

[5]



- Solid lines represent the situation when power transfer is from converter 1 to 2
- Dotted line characteristics represents reverse power flow from converter 2 to 1
- Power flow reversal is achieved by making the current order setting of converter 2 more than that of 1
- In practice, power reversal is done by swapping the current order setting between the two ends of the HVDC link

[2 marks for sketch, 1 mark for proper labelling, 2 marks for description]

b) State the typical sequence of events following a three-phase short circuit within the AC system at the rectifier end of an LCC HVDC link. Assume that the rectifier end operates in direct current control mode while the inverter ends maintain a fixed extinction advance angle before the short circuit occurs.

[5]

- Direct voltage at rectifier end and hence direct current reduces
- Current regulator decreases a to restore current until it hits the limit before switching to constant firing angle (α_{min}) mode
- Inverter switches to current control mode
- For large reduction in voltage VDCOL acts and might even have to shut the HVDC link down
- Increased consumption of reactive power at low inverter end voltages could be more detrimental to AC system than briefly shutting the DC system down

[1 mark each]

c) A thermal power plant located in close electric proximity to the rectifier end of an LCC HVDC link could be prone to torsional interaction.

i) Explain the cause of possible torsional oscillations appearing on the turbine-generator shaft.

[3]

- Torsional modes cause modulation of magnitude and phase of generator voltage
- Change in DC current / power due to modulation in AC voltage will be opposed by constant current controller at rectifier which affects the generator power (i.e. torque) output
- Phase lag between change in generator shaft speed and the corresponding torque output could become less than 90° at torsional frequency – unstable torsional oscillations due to negative damping

[1 mark each]

ii) State three factors that determine the severity of such torsional interaction.

[3]

- Rating of HVDC link with respect to the generators in the power plant
- Electric proximity between HVDC link and generators
- Presence of parallel AC lines
- Nominal firing angle

[1 mark each]

d) An LCC HVDC link is planned to interconnect two 400 kV busbars A and B of an AC system where the short circuit impedances are $Z_A = 5.5\ \Omega$ and $Z_B = 6.5\ \Omega$. The reactive power provision at both ends of the HVDC link would be 55% of its active power rating under nominal voltage.

i) If the minimum required effective short circuit ratio (ESCR) is specified as 5, determine the maximum possible active power rating (in GW) of the planned LCC HVDC link.

[5]

$$S_A = \frac{V^2}{Z_A} = \frac{400^2}{5.5} = 29.09\ \text{GVA}$$
$$S_B = \frac{V^2}{Z_B} = \frac{400^2}{6.5} = 24.62\ \text{GVA}$$

[1 mark]

$$\text{ESCR} = \frac{S - Q_c}{P_{DC}}$$

[1 mark]

$$Q_c = 0.55 \times P_{DC}$$

$$\Rightarrow P_{DC} \leq \frac{S}{\text{ESCR}+0.55}$$

[1 mark]

$$P_{DC-A} \leq \frac{S_A}{\text{ESCR}+0.55} = 5.24 \text{ GW}$$

$$P_{DC-B} \leq \frac{S_B}{\text{ESCR}+0.55} = 4.44 \text{ GW}$$

So maximum possible rating of the HVDC links is 4.44 GW

[2 marks]

ii) If a 6.0 GW LCC HVDC link were to be built between busbar A and B, what would be the ESCR at the two ends.

[2]

$$\text{ESCR}_A = \frac{S_A - 0.55 \times P_{DC}}{P_{DC}} = 4.3$$

$$\text{ESCR}_B = \frac{S_B - 0.55 \times P_{DC}}{P_{DC}} = 3.55$$

[1 mark each]

e) Explain why equidistant pulse control (EPC) is a preferred way for firing angle control for an LCC HVDC link compared to individual phase control (IPC).

[2]

- IPC relies on zero crossing of commutation voltage and is susceptible to distortions leading to harmonic instability
- For EPC the thyristors are fired independent of the zero crossing of commutation voltage. Hence EPC is more immune to harmonic problems compared to IPC.

[1 mark each]

2. a) Ideal no-load direct voltage (V_{d0}) is a simple way of expressing the average direct voltage in terms of the AC system voltage for LCC HVDC links.

i) State any two assumptions that are made towards quantifying the ideal no-load direct voltage.

[1]

1. Firing angle is considered to be zero, so the converter operates like a diode rectifier
2. AC network inductance is neglected, so no commutation overlap
3. Thyristors are considered to be ideal switches i.e. zero voltage drop when conducting and no current otherwise

[1 mark for any two points]

ii) The AC-side phase-to-neutral voltages are $e_a = E_m \cos(\omega t + 60°)$ and $e_b = E_m \cos(\omega t - 180°)$ where E_m is the amplitude of the phase-to-neutral voltages. Starting from these phase-to-neutral voltages, derive an expression for the ideal no-load direct voltage of a 6-pulse line commutated converter in terms of the RMS value of the line-to-line AC-side voltage.

[4]

Line-to-line voltage

$$e_{ab} = e_a - e_b = \sqrt{3} E_m \cos(\omega t + 30°)$$

[1 mark]

$$V_{d0} = \frac{1}{\pi/3} \int_{-\pi/3}^{0} e_{ab} d\theta$$

[1 mark]

$$= \frac{1}{\pi/3} \int_{-\pi/3}^{0} \sqrt{3} E_m \cos(\omega t + 30°)\, d\theta$$

$$= \frac{3\sqrt{3}}{\pi} E_m \cos(\omega t + 30°) \Big|_{-\pi/3}^{0} = \frac{3\sqrt{3}}{\pi} E_m = \frac{3\sqrt{6}}{\pi} E_{LN} = \frac{3\sqrt{2}}{\pi} E_{LL}$$

[2 marks]

b) A point-to-point HVDC link interconnects two separate 400 kV AC systems. The HVDC link is rated at 2200 MW and ±600 kV and uses a standard 12-pulse converter arrangement for each pole at both ends. The resistance of each pole of the HVDC line is 3.5 Ω. The AC-side

voltage at both converter stations can be assumed to be equal to the rated value. The turns ratio of the converter transformers is 0.58 with the low voltage windings on the converter side. The parameters of each 6-pulse converter used at both ends are given in Table 2.1 below:

Table 2.1: Parameters of each 6-pulse converter used at both ends of the HVDC link

Minimum firing angle	α_{min}	5°
Minimum extinction advance angle	γ_{min}	15°
Commutation resistance	R_c	6 Ω
Current margin	I_m	18%

Case I - The rectifier end of the HVDC link maintains the rated direct current while the inverter end operates with the minimum extinction advance angle. Under this situation, determine the following using the calculations on a per pole basis:

i) Direct voltage at the inverter end

[2]

Calculation on per pole basis

$B = 2, P_{d-rated} = \frac{2200}{2} = 1100$ MW, $V_{d-rated}$=600 kV, $E_{ac} = 400$ kV, $T = 0.58$

$I_d = \frac{P_{d-rated}}{V_{d-rated}} = 1.88$ kA

At inverter end

$V_{doi} = \frac{3\sqrt{2}}{\pi} E_{ac} BT = 626.62$ kV

$V_{di} = V_{doi} \cos\gamma_{min} - BR_c I_d = 583.27$ kV

[1 mark each]

ii) Firing angle (in degrees) at the rectifier

[3]

At rectifier end

$V_{dr} = V_{dr} + R_L I_d = 589.69$ kV

$V_{dor} = \frac{3\sqrt{2}}{\pi} E_{ac} BT = 626.62$ kV

$\alpha = \cos^{-1}\left[\frac{V_{dr} + BR_c I_d}{V_{dor}}\right] = 12.53°$

[1 mark each]

iii) Total reactive power drawn by the rectifier station

[2]

Power factor angle at rectifier

$$\phi_r = \cos^{-1}\left(\frac{V_{dr}}{V_{dor}}\right) = 19.77°$$

$$Q_r = 2 \times V_{dr} I_d \tan\phi_r = 777.16 \text{ MVAr}$$

[1 mark each]

iv) Total reactive power drawn by the inverter station

[2]

Power factor angle at inverter

$$\phi_i = \cos^{-1}\left(\frac{V_{di}}{V_{doi}}\right) = 21.44°$$

$$Q_i = 2 \times V_{di} I_d \tan\phi_i = 839.74 \text{ MVAr}$$

[1 mark each]

v) Commutation overlap angle (in degrees) at the rectifier

[2]

$$V_{dr} = \frac{V_{dor}}{2}[\cos\alpha + \cos(\alpha + \mu)]$$

$$\mu_r = \cos^{-1}\left(\frac{2V_{dr}}{V_{dor}} - \cos\alpha\right) - \alpha = 12.51°$$

[1 mark each]

Case II – If the AC-side voltage at the rectifier end reduces to 95% of its rated value, determine the following using the calculations on a per pole basis:

vi) Extinction advance angle (in degrees) at the inverter

[4]

At rectifier end

$$E'_{ac} = 0.95E_{ac} = 380 \text{ kV}$$

$$V'_{dor} = \frac{3\sqrt{2}}{\pi} E'_{ac} BT = 595.29 \text{ kV}$$

$$\cos\alpha = \frac{V_{dr} + BR_c I_d}{V'_{dor}} > 1$$

Hence, the rectifier cannot control current any further and the inverter takes over current control leaving the rectifier at α_{min}

[1 mark]

$$I'_d = I_d\left(1 - \frac{I_m}{100}\right) = 1.5 \text{ kA}$$

$$V'_{dr} = V'_{dor}\cos\alpha_{min} - B\,R_c I'_d = 574.98 \text{ kV}$$

[1 mark]

$$V'_{di} = V'_{dr} - R_L I'_d = 569.72 \text{ kV}$$

$$\gamma = \cos^{-1}\left[\frac{V'_{di} + BR_c I'_d}{V_{doi}}\right] = 16.34°$$

[1 mark each]

vii) Change in total reactive power drawn by the inverter station with respect to that in Case I

[3]

Power factor angle at inverter

$$\phi'_i = \cos^{-1}\left(\frac{V'_{di}}{V_{doi}}\right) = 24.6°$$

$$Q'_i = 2 \times V'_{di} I'_d \tan\phi'_i = 784.44 \text{ MVAr}$$

$$\Delta Q = Q'_i - Q_i = -55.3 \text{ MVAr}$$

[1 mark each]

viii) Commutation overlap angle (in degrees) at the inverter

[2]

$$V_{di} = \frac{V_{doi}}{2}[\cos\gamma + \cos(\gamma + \mu)]$$

$$\mu_i = \cos^{-1}\left(\frac{2V_{di}}{V_{doi}} - \cos\gamma\right) - \gamma = 14.48°$$

[1 mark each]

3. a) Answer the following questions related to the converter shown in Figure 3.1:

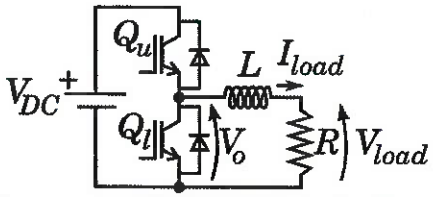
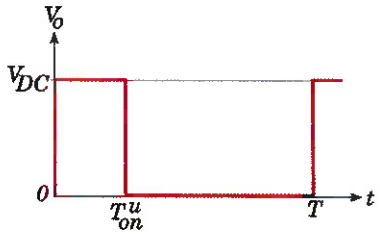


Figure 3.1 – Diagram of a DC-DC converter feeding a resistive load.

(i) Draw the waveform of the output voltage V_o over a PWM switching period. What is the relationship between the duty-cycle D of the PWM signal that drives the switches and the average of V_o (assume ideal switches)?

[2]

Assuming ideal switches (ie no voltage drop when conducting, infinite resistance when blocking and instantaneous transition between conducting and blocking), the switching sequence is the following: Q_u is left conducting for T^u_{on} time, then Q_u is blocked and Q_l is turned on for the rest of the switching period, T:



The duty cycle D is defined as: $D \triangleq \frac{T^u_{on}}{T}$. The average voltage becomes: $\hat{V}_o = DV_{DC}$

(ii) The transfer function between V_o and I_{load} is a first-order system of the form:

$$\frac{I_{load}(s)}{V_o(s)} = \frac{1}{Ls + R}$$

- What is the DC gain of this transfer function?

[1]

The DC gain can be calculated by evaluating the transfer function at s=0 and taking the magnitude of the resulting complex number. This gives a gain of $1/R$.

- What is the time constant of the system?

[1]

The time constant of the system can be found by putting the equation above in the standard form and identifying the terms:

$$\frac{I_{load}(s)}{V_o(s)} = \frac{1}{Ls + R} = \frac{\frac{1}{R}}{\frac{L}{R}s + 1} = \frac{DCgain}{\tau s + 1}$$

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This gives a time constant, τ, of L/R. Alternatively, students could identify the frequency of the pole of the transfer function and identify the time constant that corresponds to it.

- We use a PI for a closed-loop current controller that decides what voltage, V_o, the converter must output in order to track a current reference, I_{load}*. What values should K_P and K_I have in order for the closed loop transfer function to be:

$$\frac{I_{load}(s)}{I^*_{load}(s)} = \frac{1}{\tau s + 1}$$

[3]

The question suggests an inverse-based controller design, which is widely used during the course. In this case the open loop transfer function has to be of the form:

$$\mathrm{GK(s)} = \frac{1}{\tau s}$$

This means that the controller must be:

$$\mathrm{K(s)} = \frac{Ls + R}{\tau s} = \frac{\frac{L}{\tau}s + \frac{R}{\tau}}{s} = \frac{K_p s + K_i}{s}$$

Therefore, $K_p = L/\tau$ and $K_i = R/\tau$.

(iii) Consider now the case where this converter is connected to a DC grid represented by a DC voltage source as shown in Figure 3.2:

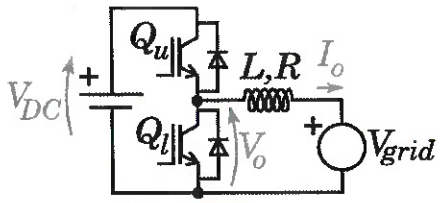


Figure 3.2 – Diagram of a DC-DC converter connected to a DC grid represented by a voltage source.

- What equation would you use to calculate the value of the current reference I_o* for a given power output reference value P*? Explain what variables you would need to measure in real time in order to be able to perform the calculation in the controller.

[2]

Typically, we would assume that the losses of the output filter are negligible and the variation of the energy stored in the inductor is negligible as well. This would imply that the instantaneous power calculated over Vo would be the same as the one calculated over Vgrid. The advantage would be that Vgrid is set by the network and can be assumed to be approximately constant if the grid is strong (or the converter is comparatively small). Therefore, the current reference can be calculated as:

$$I_o^* := \frac{P^*}{V_{grid}}$$

This requires Vgrid to be measured in real time. Alternatively, Vgrid could be assumed to be constant given that this is a DC application; however, in that case there would be no guarantee that the calculated current reference would yield the correct power output.

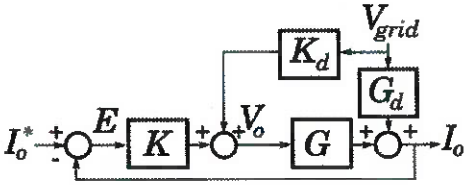
- Draw a block diagram showing how V_o and V_{grid} affect I_o. Show where the closed-loop current control and the feed-forward controller would appear in the diagram.

[3]

Unlike the case in questions ii and ii, the the current is affected both by the voltage out the converter and the voltage of the grid. This can be written as:

$$Io(s) = G(s)Vo(s) + G_d(s)V_{grid}(s)$$

The voltage applied by the converter, as suggested by the text in the question can be chosen using a combination of a feedback controller, $K(s)$, fed with the error between output and reference plus a feedforward controller, $K_d(s)$, that measures the disturbance (V_{grid}) and calculates the control action that ideally would cancel out the effect of the disturbance. The diagram of the system including both controllers is shown here:



b) Answer the following questions related to the single phase inverter shown in Figure 3.3:

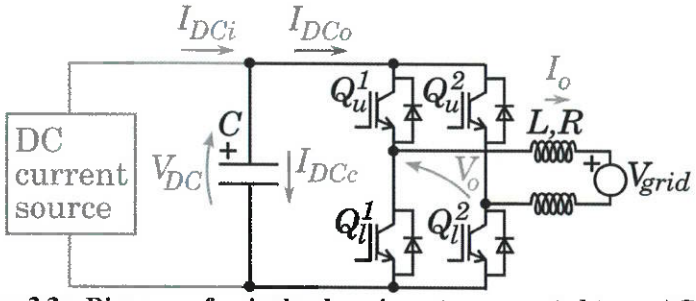


Figure 3.3 – Diagram of a single phase inverter connected to an AC grid

Parameter	Value	Units	Brief description
V_{grid}	230	V rms	Rated AC grid voltage
P_{rated}	1	kW	Rated active power delivered to the grid
f_{rated}	50	Hz	Rated frequency of the AC grid
R	0.25	Ω	Equivalent resistance of bridge plus the filter
L	15	mH	Inductance of the filter

Table 3.1 – Diagram of a single phase inverter connected

(i) Calculate the base impedance given the converter characteristics and calculate the impedance of the output filter in per-unit.

[2]

The first step is to calculate the base impedance, which gives:

$$Z_{base} = \frac{(V_{grid})^2}{P_{rated}} = 52.9\ \Omega$$

Once this is known, we can calculate all other impedances in pu as the ratio between their actual value in Ohms and the magnitude of the base impedance:

$$R^{pu} = \frac{0.25}{52.9} = 0.47\%$$

and

$$X^{pu} = \frac{\omega_e L}{Z_{base}} = \frac{2\pi 50\ 15\ 10^{-3}}{52.9} = 8.9\%$$

(ii) What is the capacitance of the DC link if the DC link voltage V_{DC} is of 400 V and the inertia time of the DC link, H, is of 50 ms?

[2]

The inertia time, H, is defined as the ratio between the energy stored and the rated power of the device. Using the expression of the energy of a capacitor and solving for the capacitance this gives:

$$C = \frac{2\ H\ P_{rated}}{(V_{DC})^2} = 0.625\ mF$$

(iii) Consider the operating point where the converter is delivering rated power to the AC grid with unity power factor (ie P_{grid}=P_{rated}, Q_{grid}=0). Answer the following questions:

- What is the output current, I_o, and the voltage, V_o, the converter must apply?

[1]

Knowing the voltage of the grid, V_{grid}, the output current can be calculated as:

$$I_o := \frac{P_{rated}}{V_{grid}} = 4.35\ A$$

This current would be in phase with Vgrid given that the reactive power delivered is zero. On the other hand, the voltage the converter would apply is:

$$V_o = V_{grid} + (R + i\omega_e L)I_o = 232\ V \angle 5^o$$

- What are the conduction losses of the converter? How much reactive power does the filter consume?

[1]

The conduction losses can be calculated as the aggregate resistance multiplied by the square of the output current:

$$P_{loss} = R(I_o)^2 = 4.73\ W$$

Similarly, the reactive power of the filter is:

$$Q_{filter} = \omega_e L(I_o)^2 = 90\ W$$

(iv) Consider now the operating point where the converter is delivering 0 apparent power to the grid. What voltage, V_o, does the converter need to apply? What would the minimum voltage of the DC link V_{DC} be in order to be able to generate such voltage?

[3]

In this situation the output current is zero and the output voltage of the converter matches the voltage of the grid, ie V_o=V_{grid}=230 V (rms).

The DC link voltage must be high enough for the converter to modulate the required output voltage. If we want this voltage to be modulated without low frequency harmonic distortion for a full period of the AC grid frequency, the converter can generate any voltage in the range between -V_{DC} and V_{DC}. Therefore, V_{DC} must be greater or equal to the peak voltage of the AC voltage, which gives $V_{DC} \geq 325$ V.

c) Consider a basic one-arm modular converter connected between an AC grid and a DC grid represented as equivalent voltage sources as shown in Figure 3.4 (its parameters are summarised in Table 3.2). Answer the questions below:

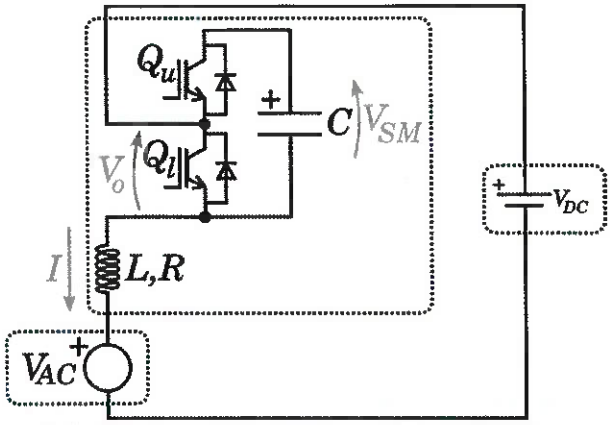


Figure 3.4 – Diagram of a one-arm modular AC/DC converter

Parameter	Value	Units	Brief description
V_{DC}	750	V	Rated DC grid voltage.
V_{AC}	398	V rms	Rated AC grid voltage
f_{rated}	50	Hz	Rated frequency of the AC gr
R	10	mΩ	Equivalent resistance of bridge plus the filter
L	0.34	mH	Inductance of the filter

Table 3.2 – Diagram of a single phase inverter connected

(i) Calculate the AC and DC components of the output current, I, and the output voltage of the arm, V_o, for a stable operating point (0 net power balance across the submodule) when the power delivered to the AC grid is P_{AC}=100 kW, Q_{AC}=0 kVAr.

[4]

The condition of 0 net power balance, implies that the power exchanged by the submodule through its AC voltage and current is equal to the power exchanged through its DC voltage and current with opposite sign. The first step to solve this problem is to find out what the AC current is going to be based on the setpooint of the converter:

$$I_o^{AC} = \frac{P_{AC}}{V_{AC}} = 251\ \text{A}$$

In this case, the reactive power is zero and the AC output current will be aligned with the voltage of the AC grid.

The AC output voltage of the submodule is:

$$V_o^{AC} = -V_{AC} - (R + i\omega_e L) I_o^{AC} = 401\ \text{V} \angle -176º$$

Next, we can calculate the active power the submodule will exchange using AC current, this is:

$$P_o^{AC} = \text{Re}\{V_o^{AC} (I_o^{AC})^*\} = -100.53\ \text{kW}$$

Notice that this power is greater due to the voltage drop caused by the resistance of the circuit.

This power must be matched by the power the submodule exchanges using AC and DC current, that is:

$$P_o^{DC} = -P_o^{AC} = 100.53\,\text{kW}$$

Next we will calculate the DC voltage and the DC current the submodule must apply. We know that:

$$V_{DC} - V_o^{DC} = RI_o^{DC}$$

and

$$P_o^{DC} = V_o^{DC} I_o^{DC}$$

This leads to the following solution:

$$V_o^{DC} = \frac{V_{DC} + \sqrt{(V_{DC})^2 - 4\,RP_o^{DC}}}{2} = 748.7\,\text{V}$$

and

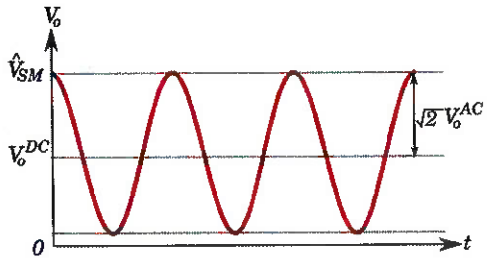
$$I_o^{DC} = \frac{P_o^{DC}}{V_o^{DC}} = 134\,\text{A}$$

[This question includes multiple steps, marks will be given if for example students are able to solve the AC variables but don't calculate the DC variables. Full marks will require the complete solution; full marks will be given if students neglect the losses of the resistance when calculating the DC current only if they state the assumption and they provide enough insight in their explanation for us to confirm that they understand the additional complexity involved in not doing such simplification.]

(ii) Draw the approximate waveform of the output voltage, V_o, at zero power (I=0). Explain how high V_{SM} must be and what the ratio between V_{DC} and V_{AC} must be for the converter to be able to operate under this specific condition.

[3]

At zero power, the converter will apply the same voltage as the AC grid (with opposite sign) and the same voltage and the DC grid as the voltage drop caused by current will be zero. Therefore, the voltage will look like:



Firstly, the submodule voltage must be greater or equal to the summation of the DC voltage plus the peak AC voltage in order for the submodule to be able to generate the required voltage. This implies:

$$V_{SM} \geq \hat{V}_{SM} = V_o^{DC} + \sqrt{2}V_o^{AC}$$

Secondly, the converter uses a half-bridge submodule topology, this means that the output voltage must be greater than zero, this implies that:

$$V_o^{DC} \geq \sqrt{2} V_o^{AC}$$

Which is equivalent to:

$$\frac{V_o^{DC}}{V_o^{AC}} \geq \sqrt{2}$$

(iii) List the potential advantages and disadvantages of designing this converter to use three submodules in the arm compared to using a single submodule.

[2]

Potential advantages:

- With three submodules, the voltage rating of each switch is divided by three, which enables designing the converter for high DC voltages (high power).
- With three submodules, phase-shifting can be used to reduce the harmonic content of the generated voltage while decreasing the switching frequency of the individual switches, which can reduce the filtering requirements of the arm inductor and reduce the switching losses.
- With three submodules, the converter can be designed for redundancy, for example by designing the converter to be able to operate using just two submodules.

Potential disadvantages:

- With three submodules, the number of switches in the conduction path is multiplied by three. If the same devices were used, the conduction losses could increase by a factor of three.
- With three submodules, the number of auxiliary circuits in the converter increases by a factor of three (e.g. three submodule bridge drivers, voltage measurements, etc).
- With three submodules, the volume of the converter is likely to be greater than it would be for an equivalent single submodule converter.
- With three submodules, the complexity of the control system and the number of sensors required is greater. The difference would be more significant the greater the number of submodules.